

# FQCNN: A Fully Quantum-Native Convolutional Neural Network with Coherent Unitary Pooling

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**Abstract**—Quantum convolutional neural networks (QCNNs) commonly reduce registers through mid-circuit measurement or classical feed-forward, while fully coherent alternatives are difficult to compare fairly. We study a fully quantum-native convolutional neural network (FQCNN) that performs state preparation, shared register-index convolution, pooling, and classification with unitary operations before one terminal observable. The ten-qubit model uses global amplitude encoding, shared three-axis single-qubit rotations, and a measurement-free pooling block. We prove that the stated block induces exactly the corresponding measure-and-condition channel on the retained register, without mid-circuit measurement, reset, or classical feed-forward. The validated evaluation covers four binary tasks from three dataset families, five random seeds, identical split manifests, logistic and dense-neural baselines, and a tree-tensor baseline. The FQCNN is below the logistic and dense-neural baselines in mean accuracy on all four tasks and above the tree-tensor baseline on all four; no primary paired difference is significant after Holm correction. Pooling-transfer, finite trainability, resource, and local-noise checks provide bounded supplementary evidence. All results are classical simulation or local fake-backend checks; no quantum advantage or hardware-performance claim is made.

**Index Terms**—Amplitude encoding, quantum convolutional neural networks, quantum machine learning, unitary pooling.

## I. INTRODUCTION

Quantum machine learning has become an active research area that studies how quantum computing can be applied to data-driven, learning-based tasks. One recurring interest is how to bring useful inductive biases, such as locality, parameter sharing, and hierarchical feature extraction, into the quantum setting. Among classical models, the convolutional neural network has been especially successful because it exploits spatial structure in data, which motivated quantum convolutional neural networks (QCNNs) that aim to carry the principles of classical CNNs into quantum circuits [10].

Several approaches have extended convolutional ideas to the quantum domain through structured variational circuits, usually referred to as QCNNs. These models tend to rely on hierarchical constructions that apply local quantum operations followed by dimensionality reduction [10], [13]. In the canonical construction, pooling is implemented through measurement-conditioned operations, which introduces a mid-circuit measurement and classical-control boundary [10]. The present work isolates that boundary and tests a coherent alternative; it does not claim that every QCNN variant has the same implementation.

When a pooling construction uses measurement-conditioned corrections, its implementation requires mid-circuit measurement and classical control. The availability, latency, and calibration cost of those operations are platform dependent, so their removal is an architectural distinction rather than, by itself, evidence of a hardware speedup. Deep or highly connected variational circuits can also accumulate noise and face difficult optimisation landscapes, but those effects must be measured for a specified circuit and device rather than inferred from an architecture label alone.

This motivates two bounded questions. First, can the stated measure-and-condition pooling channel be represented by a coherent unitary block without dynamic control? Second, how does one frozen implementation behave under matched data splits, finite simulator experiments, and explicit local-noise and resource protocols? We answer those questions without claiming that the resulting circuit is universally trainable, noise robust, or ready for physical deployment.

In this work we address these challenges by introducing a fully quantum-native neural network (FQCNN) architecture designed to preserve coherence across every stage of the learning pipeline. The model performs data encoding, convolution, pooling, and classification entirely through unitary operations, removing the need for intermediate measurement or classical

post-processing. The quantum convolution uses a shared  $2 \times 2$  kernel over register-index windows, which controls parameter growth while keeping circuit depth low; global amplitude encoding means that this is not an image-space translational-equivariance guarantee. Dimensionality reduction is achieved through a coherent pooling mechanism that uses unitaries to compress information while keeping the global state pure. Section IV-F proves that this mechanism reproduces measurement-based pooling exactly, so the benefit it delivers is the removal of mid-circuit control requirements rather than the retention of otherwise-lost information.

Beyond its architectural construction, the study asks how the frozen model behaves under a reproducible multi-domain protocol. The historical MNIST single-seed result is retained as provenance; the primary aggregate supplies the performance estimates and uncertainty used for the final claims. The present work establishes simulator-level architectural consistency, not physical-device performance.

The main contributions of this work are as follows:

- 1) We propose a fully quantum-native convolutional neural network that maintains coherence across encoding, convolution, pooling, and classification, with no intermediate disruption or classical post-processing.
- 2) We introduce a parameter-shared quantum convolutional kernel with bounded register-index gate support; global amplitude encoding is kept distinct from image-space locality.
- 3) We develop a coherent unitary pooling mechanism that performs dimensionality reduction through quantum operations alone.
- 4) We evaluate the frozen model with same-split classical and quantum baselines across four binary tasks from three dataset families, reporting uncertainty and negative results.

The rest of the paper is organised as follows. Section II reviews related work on quantum convolutional and variational quantum models. Section III defines the evaluation questions and evidence boundaries. Section IV describes the proposed FQCNN architecture and its mathematical formulation. Section V covers training and optimisation, and Section VI reports the experimental results and analysis. Section VII discusses the architectural and empirical implications and states the limitations and future work. Section VIII records the novelty and technical differentiation, followed by data and code availability and the conclusion.

## II. RELATED WORK

### A. Quantum Convolutional Neural Networks

Quantum convolutional neural networks extend classical convolutional networks into the quantum domain by combining local unitary operations with hierarchical dimensionality reduction. The canonical QCNN uses a structured variational circuit inspired by renormalisation techniques [10], while hierarchical quantum classifiers provide a related tree-like construction [13]. These papers establish the architectural

context for the present comparison; detailed claims about hardware-aware variants and cross-paper benchmark rankings are intentionally outside this manuscript’s evidence base. In particular, measurement-based pooling requires a mid-circuit measurement and classical feed-forward, whereas the proposed block keeps the register coherent until the terminal observable [10].

### B. Variational Quantum Classifiers and Trainability

The broader framework of variational quantum circuits (VQCs) underlies most quantum machine learning models [12]. Optimisation commonly uses gradient-based updates and stochastic variants [2], [5]. Increasing circuit depth and expressibility does not always help; it can produce barren plateaus, where gradient magnitudes become small as system size grows [1], [4], [11]. Strategies based on structured ansatz design and cost-function-aware optimisation can soften these effects [14]. These observations motivate empirical trainability checks for each proposed architecture; they do not establish a universal scaling law.

### C. Quantum Autoencoders and Information Compression

Dimensionality reduction in quantum models is often expressed through quantum autoencoder frameworks, which compress quantum states into lower-dimensional subspaces while preserving relevant information [8], [9]. Recent journal work also studies information compression with structured quantum autoencoders [7]. Autoencoders offer a principled route to quantum information compression, but they typically require separate encoding–decoding stages. By contrast, embedded pooling mechanisms that operate directly within QCNN architectures provide an alternative path to information reduction.

### D. Noise and Error Mitigation in Quantum Learning

Noise remains a primary obstacle to deploying quantum machine learning models on near-term hardware. Comprehensive reviews of quantum error mitigation cover strategies such as zero-noise extrapolation, data-driven correction, and probabilistic error cancellation [6]. Recent work also studies data-augmented learning for noise mitigation [3]. These methods operate at the error-mitigation or training layer; the present manuscript instead treats architectural noise behaviour as a specified finite-simulation question.

## III. EVALUATION QUESTIONS AND EVIDENCE BOUNDARIES

The evaluation is organised around four questions that separate an architectural identity from an empirical performance claim. This separation is important for a fully coherent model: a circuit can remove a dynamic-control requirement without improving accuracy, and an accuracy result can be useful as a benchmark without establishing a quantum advantage. The questions are therefore evaluated with different units of evidence rather than collapsed into one score.

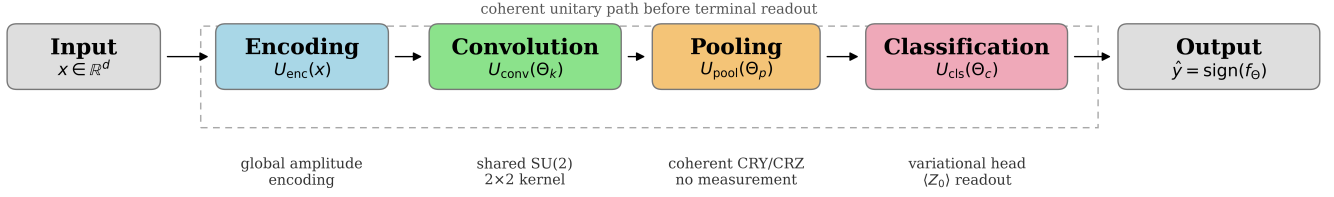


Fig. 1. Overall FQCNN architecture. The pipeline runs Input  $\rightarrow$  Encoding  $\rightarrow$  Convolution  $\rightarrow$  Pooling  $\rightarrow$  Classification  $\rightarrow$  Output. Every stage operates through unitary quantum operations, with no intermediate measurements.

TABLE I  
EVALUATION QUESTIONS, EVIDENCE LAYERS, AND THE CORRESPONDING CLAIM BOUNDARY. THE TABLE DISTINGUISHES EXACT STATEMENTS ABOUT THE IMPLEMENTED CIRCUIT FROM FINITE EMPIRICAL OBSERVATIONS.

| Question                                              | Operational test                                                                                                                            | Evidence used                                                                                 | Interpretation permitted                                                                               |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Does the main path remain coherent?                   | Inspect the frozen circuit signature and terminal measurement; verify that intermediate operations are unitary.                             | Canonical circuit source, architecture fixtures, and the state-purity audit.                  | A property of the frozen implementation; not a hardware execution claim.                               |
| Does coherent pooling reproduce the compared channel? | Compare the controlled unitary followed by a partial trace with the explicitly constructed measure-and-condition channel.                   | Theorem 1, E1 channel comparison, and E2 dephasing controls.                                  | Exact equivalence for the stated block, with no claim that information or accuracy is recovered.       |
| How does the frozen model behave empirically?         | Use predeclared binary tasks, common ordered samples, common split manifests, five seeds, and one final test evaluation.                    | Four-task Q1 aggregate, same-split logistic/MLP/TTN baselines, uncertainty, and paired tests. | A finite within-study comparison; no state-of-the-art or quantum-advantage claim.                      |
| What limits practical interpretation?                 | Audit effective capacity, finite-size gradients, expressibility, information dynamics, noise sensitivity, and logical/transpiled resources. | Versioned model-analysis and local Qiskit evidence, each with its own protocol.               | Bounded simulator and local-transpilation observations; no asymptotic or device-performance guarantee. |

#### A. From architectural identity to empirical estimands

The first two questions are principally mathematical. The statement that the main path is unitary follows from the operation types in the executed tape, and the pooling statement follows from the controlled-unitary factorisation proved in Section IV-F. Their numerical checks are regression tests at the frozen headline configuration rather than estimates of a population. This distinction prevents a finite floating-point residual from being mistaken for an empirical approximation to a theorem.

The performance question has a different estimand. For task  $d$  and model arm  $m$ , let  $A_{d,m,s}$  be the accuracy on the final test partition for seed  $s$ . The reported point estimate is the mean over the registered seed set,

$$\begin{aligned} \bar{A}_{d,m} &= \frac{1}{S} \sum_{s=1}^S A_{d,m,s}, \\ s_{d,m} &= \left[ \frac{1}{S} \sum_{s=1}^S (A_{d,m,s} - \bar{A}_{d,m})^2 \right]^{1/2}. \end{aligned} \quad (1)$$

where  $S = 5$  in the primary matrix. The standard deviation in the tables is the population standard deviation of these five registered runs, not a claim that the five seeds are independent draws from all possible training procedures. For a comparator

$c$ , the paired effect is computed seed by seed as  $\Delta_{d,c,s} = A_{d,\text{FQCNN},s} - A_{d,c,s}$ . This makes the split and seed the unit of the primary comparison and keeps per-example tests in a descriptive role.

The third question is consequently not phrased as “is the FQCNN better?” Instead, it asks where the fixed architecture sits relative to transparent baselines under one reproducible protocol. This choice is deliberate. The model has a globally amplitude-encoded input and only ten qubits, while the classical baselines see the original 784 flattened features. The fair comparison is therefore same data identity, same split identity, same seed set, and the same validation-only selection rule; it is not an assertion of equal parameter count or equal computational cost.

#### B. Evidence hierarchy and negative findings

The evidence package is layered so that a failed or null result at one layer does not get silently replaced by a stronger claim at another layer. Exact channel equivalence is not converted into a claim of better information retention. A non-significant paired test is not converted into equivalence with five pairs. A finite gradient sweep is not converted into an asymptotic trainability theorem. Similarly, local fake-backend transpilation is not called hardware validation. This hierarchy is also why the paper retains the negative findings:

the FQCNN is below the logistic and dense baselines on all four primary tasks, the wider coherent and SU(4) pooling families do not automatically improve the frozen construction, and the full dynamical Lie algebra does not supply the planned polynomial-size trainability certificate.

These are not discarded intermediate observations. They delimit what the architectural contribution can support and make the paper useful as a checked reference implementation rather than a selective benchmark report. All quantities introduced in the remainder of the paper are either algebraic consequences of the circuit or are linked to a retained machine-readable artifact in the availability and reproducibility records.

#### IV. PROPOSED FQCNN ARCHITECTURE

##### A. Design requirements and register notation

The architecture was frozen against four requirements. First, all trainable transformations after state preparation must be expressible as quantum gates with a single terminal observable. Second, the convolutional kernel must be shared across its register-index windows so that parameter allocation is independent of the number of windows. Third, pooling must reduce the active register without inserting a measurement or a classical branch. Fourth, the implementation must expose enough structure to audit dead parameter slots, discarded wires, and resource costs separately.

Let  $A_0 = \{0, \dots, n-1\}$  denote the active wires immediately after state preparation. At pooling stage  $\ell$ , the executed pairing is  $P_\ell = \{(a_i, b_i)\}_{i=1}^{m_\ell}$ , where  $a_i$  is retained and  $b_i$  is compressed. The retained and compressed sets are  $R_\ell = \{a_i\}$  and  $C_\ell = \{b_i\}$ . If the active-register cardinality is odd, the final unpaired wire is collected in  $O_\ell$  and retired at the same stage. Writing  $U_\ell$  for the convolution and pooling operations emitted at that stage, the state passed to the next active register is represented by

$$\rho_{\ell+1} = \text{Tr}_{C_\ell \cup O_\ell} [U_\ell \rho_\ell U_\ell^\dagger], \quad A_{\ell+1} = R_\ell. \quad (2)$$

Equation (2) is a description of the reduced state on the wires used later; the full state remains pure before the inactive wires are ignored. No reset is inserted into the headline circuit. The partial trace is the mathematical description of the fact that a retired wire is not addressed by later gates or the terminal observable.

The headline schedule is not a conventional square image lattice after the first reduction: ten wires become five, then two, then one. The convolutional window generator therefore operates on the current active-index arrangement, whereas the input itself remains a global amplitude state. This is the reason the paper uses “register-index window” throughout. It describes the actual gate support while avoiding an unsupported claim of pixel-space translation equivariance.

##### B. Problem Formulation

Consider a labelled dataset  $\mathcal{D} = \{(x^{(k)}, y^{(k)})\}$  for  $k = 1, \dots, N$  in a supervised setting, where  $x \in \mathbb{R}^d$  is the input feature vector,  $x_i \in [-1, 1]$ ,  $y \in \{-1, 1\}$ , and  $d$  is the

number of features after preprocessing. To scale efficiently, we use amplitude encoding, which maps the input vector into a quantum state. We consider a register of  $n = 10$  qubits all initialised in the ground state

$$|\psi_0\rangle = |0\rangle^{\otimes 10}. \quad (3)$$

A two-dimensional input image  $X \in \mathbb{R}^{28 \times 28}$  is flattened into a 784-dimensional vector  $x$ , which is zero-padded to 1024 and mapped to the  $2^{10} = 1024$  state amplitudes using global amplitude encoding. The full model is a parameterised unitary circuit built from the following operations:

$$U(\Theta; x) = U_{\text{cls}}(\theta_c) U_{\text{pool}}(\theta_p) U_{\text{conv}}(\theta_k) U_{\text{enc}}(x), \quad (4)$$

where  $\Theta = \{\theta_c, \theta_p, \theta_k\}$ . The scalar model output is defined as

$$f_\theta(x) = \langle \psi_0 | U^\dagger(\Theta; x) Z_0 U(\Theta; x) | \psi_0 \rangle, \quad (5)$$

and classification is performed via

$$\hat{y}(x) = \text{sign}(f_\theta(x)). \quad (6)$$

All intermediate transformations are strictly unitary, with no mid-circuit measurements.

##### C. Quantum Data Encoding

The frozen implementation uses one state-preparation operation: global amplitude encoding. It maps a normalised classical input vector into the amplitudes of a ten-qubit state, representing up to  $2^{10}$  features without introducing a second data-dependent feature map. Prior to encoding, each input feature is min-max normalised to the bounded domain  $[a, b] = [0, 1]$ :

$$x'_i = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}} \cdot (b - a) + a, \quad [a, b] = [0, 1]. \quad (7)$$

The preprocessed vector is padded with zeros to length  $2^{10}$  and prepared as

$$|\psi_{\text{enc}}(x)\rangle = \text{AmplitudeEmbedding}(x', \text{wires} = 0:9, \text{normalize} = \text{true}, \text{pad} = 0) |0\rangle^{\otimes 10}. \quad (8)$$

No per-qubit rotations or data-dependent ring-correlation gates are applied on top of this operation in the main amplitude-encoded path. Consequently, the locality statements below refer to the support and parameter sharing of variational gates in the register-index space, not to spatially adjacent pixels in the original image.

##### D. Quantum Convolutional Layer

The convolutional layer applies a parameter-shared variational kernel over all  $2 \times 2$  sliding windows. The complete pipeline is summarised in Fig. 1; the circuit-level kernel used in each window is shown in Fig. 2.

TABLE II

EXECUTED REGISTER SCHEDULES AND PARAMETER ALLOCATION USED BY THE FINITE-SIZE AUDITS. “ALLOCATED” INCLUDES SLOTS RESERVED BY THE MODEL LAYOUT; “EFFECTIVE” COUNTS SLOTS WITH A GRADIENT MAGNITUDE ABOVE  $10^{-12}$  IN THE CORRESPONDING FINITE DIAGNOSTIC.

| $n$ | Input geometry | Active-wire schedule                           | Pooling stages | Allocated slots | Effective slots |
|-----|----------------|------------------------------------------------|----------------|-----------------|-----------------|
| 4   | $4 \times 4$   | $4 \rightarrow 2 \rightarrow 1$                | 2              | 188             | 62              |
| 6   | $8 \times 8$   | $6 \rightarrow 3 \rightarrow 1$                | 2              | 194             | 62              |
| 8   | $16 \times 16$ | $8 \rightarrow 4 \rightarrow 2 \rightarrow 1$  | 3              | 260             | 122             |
| 10  | $28 \times 28$ | $10 \rightarrow 5 \rightarrow 2 \rightarrow 1$ | 3              | 269             | 74              |

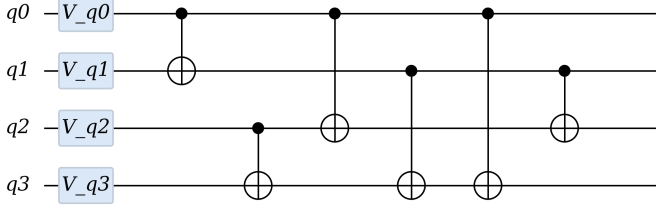


Fig. 2. The  $2 \times 2$  register-index convolutional kernel acting on window  $\omega = \{q_0, q_1, q_2, q_3\}$ . The gates  $V_q$  apply full  $SU(2)$  rotations per qubit with parameters shared across all register-index windows. The six CNOTs realise all-to-all entanglement inside the window; this is a gate-support statement, not an image-space equivariance claim.

1) *Kernel parameterisation*: Let each window contain four qubits. For a kernel depth  $d$  and depth slice  $k$ , the single-qubit rotation block is

$$V_q^{(k)}(\Theta_k) = R_X(\theta_{X,q}^{(k)}) R_Z(\theta_{Z,q}^{(k)}) R_Y(\theta_{Y,q}^{(k)}). \quad (9)$$

Each qubit is parameterised using full  $SU(2)$  rotations in the selected ansatz while keeping the circuit shallow. Here  $\Theta_k \in \mathbb{R}^{4 \times d \times 3}$ , so  $|\Theta_k| = 12d$ . Parameter sharing enforces  $\Theta_k(\omega) = \Theta_k$  for all register-index windows  $\omega$ , so the parameter count does not grow with the number of windows. Because the input is globally amplitude encoded, this sharing is not an image-space translational-equivariance guarantee.

2) *Intra-window entanglement*: Within each window, entanglement is introduced through the unitary

$$U_{\text{ent}}(\omega) = \text{CNOT}_{q_1 \rightarrow q_2} \text{CNOT}_{q_0 \rightarrow q_3} \text{CNOT}_{q_1 \rightarrow q_3} \\ \times \text{CNOT}_{q_0 \rightarrow q_2} \text{CNOT}_{q_2 \rightarrow q_3} \text{CNOT}_{q_0 \rightarrow q_1}, \quad (10)$$

with the rightmost operator acting first. The six gates couple every pair of qubits in the register-index window. This all-to-all pattern provides bounded gate support at a fixed depth, while entanglement does not cross a window boundary. We report this as a structural property of the implemented register topology; it is not a proof of image-space locality or asymptotic trainability. The full convolution operator is

$$U_{\text{conv}}(\Theta_k) = \prod_{\omega} U_{\text{ent}}(\omega) V_{\omega}(\Theta_k). \quad (11)$$

### E. Coherence-Preserving Pooling

1) *Pooling pair set*: After convolution, the qubits are grouped into disjoint ordered pairs as illustrated in Fig. 3.

$$P \subseteq \{(a, b) \mid a, b \in \{1, \dots, N\}, a \neq b\}. \quad (12)$$

Each pair  $(a, b) \in P$  designates  $a$  as the retained qubit and  $b$  as the compressed qubit, with pairs mutually disjoint:

$$(a, b), (c, d) \in P \Rightarrow \{a, b\} \cap \{c, d\} = \emptyset. \quad (13)$$

2) *Pooling unitary*: For every pair  $(a, b) \in P$ , with the compressed qubit  $b$  acting as control and the retained qubit  $a$  as target, we apply

$$W_{ab}(\Theta_p) = R_Y^{(a)}(\gamma_{ab}) R_Y^{(b)}(\epsilon) \text{CRZ}_{b \rightarrow a}(\beta_{ab}) \text{CRY}_{b \rightarrow a}(\alpha_{ab}), \quad (14)$$

where operators act right to left and  $\epsilon = 0.02$  is a fixed (non-trainable) angle applied to the compressed qubit. Because  $b$  is excluded from all subsequent operations,  $R_Y^{(b)}(\epsilon)$  has no effect on any observable of the retained register; it is retained here only because the reported parameter vector and its random initialisation stream are those of the executed model. Section IV-F makes this inertness precise. The global pooling operator is

$$U_{\text{pool}}(\Theta_p) = \prod_{(a,b) \in P} W_{ab}(\Theta_p). \quad (15)$$

The control assignment in (14) is not incidental: both two-qubit gates are diagonal in the computational basis of the compressed qubit  $b$ , which is the structural property Section IV-F rests on.

3) *Dimensional reduction*: After pooling, only the retained qubits  $a$  stay active; the compressed qubits  $b$  are discarded after the unitary evolution. At a stage with  $N$  active wires, the executed pair-and-retire schedule forms  $m = |P| = \lfloor N/2 \rfloor$  consecutive pairs and retires any unpaired wire. Thus the active output count is  $n_r = m$  (rather than  $N - m$  when  $N$  is odd), and the pooling parameter count is  $|\Theta_p| = 3|P|$ . For the ten-qubit model the active-wire schedule is  $10 \rightarrow 5 \rightarrow 2 \rightarrow 1$ . The pooling therefore performs structured dimensionality reduction while preserving the pre-discard global state as a pure state.

### F. Unitary Versus Measurement-Based Pooling

A measurement-based QCNN pooling block can measure the compressed qubit and apply a classically conditioned correction to the retained qubit. The construction here instead uses a coherent two-qubit unitary. This section establishes the exact relationship between those two specified blocks: the block in (14) reproduces the stated measurement-based channel exactly, without requiring mid-circuit measurement, qubit reset, or classical feed-forward.

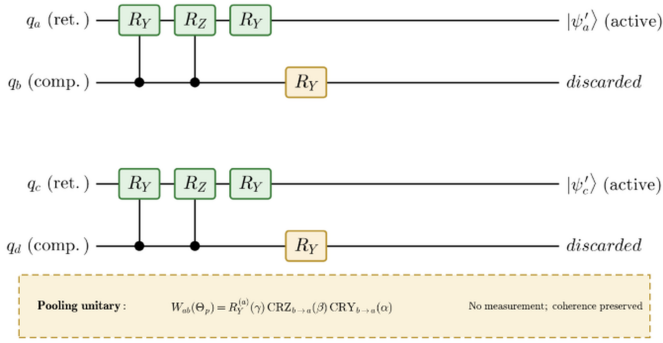


Fig. 3. Coherent unitary pooling mechanism. For each disjoint pair  $(a, b) \in P$ , qubit  $a$  is retained and  $b$  is compressed. The unitary  $W_{ab}$  transfers information from  $q_b$  into  $q_a$  via CRY and CRZ gates *controlled on*  $q_b$  and targeting  $q_a$ , followed by a local  $R_Y$  on each wire, with no mid-circuit measurement. The compressed qubit is excluded from all later operations; any unpaired wire is also retired by the executed schedule, while the global state remains pure before the active-register reduction.

1) *Setup*: Fix a pair  $(a, b) \in P$  and let  $\rho$  be the joint state of  $(a, b)$  immediately before pooling. A *measure-and-condition* pooling step measures  $b$  in the computational basis and, on outcome  $m \in \{0, 1\}$ , applies  $U_m \in SU(2)$  to  $a$ . Averaged over outcomes, the retained register undergoes the channel

$$\mathcal{M}_{\{U_m\}}(\rho) = \sum_{m \in \{0, 1\}} U_m \langle m|_b \rho |m\rangle_b U_m^\dagger. \quad (16)$$

**Theorem 1** (Exact simulation). *Let  $U_0 = R_Y(\gamma_{ab})$  and  $U_1 = R_Y(\gamma_{ab})R_Z(\beta_{ab})R_Y(\alpha_{ab})$ . Then for every input state  $\rho$ ,*

$$\text{Tr}_b[W_{ab}\rho W_{ab}^\dagger] = \mathcal{M}_{\{U_0, U_1\}}(\rho). \quad (17)$$

*That is, the coherent block in (14) induces exactly the measure-and-condition channel (16) on the retained register, and therefore reproduces every observable statistic of measurement-based pooling.*

*Proof.* Both two-qubit gates in (14) are controlled on  $b$ , hence diagonal in  $b$ 's computational basis:

$$\text{CRZ}_{b \rightarrow a}(\beta) \text{CRY}_{b \rightarrow a}(\alpha) = \sum_m |m\rangle\langle m|_b \otimes V_m, \quad (18)$$

with  $V_0 = I$  and  $V_1 = R_Z(\beta)R_Y(\alpha)$ . The local  $R_Y^{(a)}(\gamma)$  commutes with every projector on  $b$ , so absorbing it gives  $|m\rangle\langle m|_b \otimes U_m$  with  $U_m = R_Y(\gamma)V_m$ , matching the statement. Writing  $V = \sum_m |m\rangle\langle m|_b \otimes U_m$  we have  $W_{ab} = (R_Y^{(b)}(\epsilon) \otimes I_a)V$ , and since the partial trace over  $b$  is invariant under any unitary acting on  $b$  alone,

$$\text{Tr}_b[W_{ab}\rho W_{ab}^\dagger] = \text{Tr}_b[V\rho V^\dagger] = \sum_m U_m \langle m|_b \rho |m\rangle_b U_m^\dagger. \quad (19)$$

□

The same result can be written in Kraus form, which makes the deferred measurement interpretation explicit. Define

$$K_m = U_m |m\rangle_b, \quad m \in \{0, 1\}. \quad (20)$$

The operators map the joint pair to the retained subsystem and satisfy  $\sum_m K_m^\dagger K_m = I_a \otimes I_b$ . Hence  $\mathcal{M}_{\{U_m\}}(\rho) = \sum_m K_m \rho K_m^\dagger$  is a completely positive trace-preserving map. The coherent implementation replaces the classical record  $m$  with the computational-basis value already present in the control wire. Tracing that wire at the end of the pooling block produces the same Kraus sum because the partial trace contracts the two copies of  $|m\rangle\langle m|$  and removes cross terms between distinct outcomes.

This derivation also identifies the exact boundary of the result. The compressed wire's off-diagonal blocks are not used by the controlled block, so they cannot influence the retained output after tracing out that wire. A general two-qubit unitary need not have this property: it can rotate the compressed subsystem before the reduction and thereby make its coherences operationally visible. The strict-containment proposition is therefore a statement about the available channel family, not a claim that the frozen three-angle block exploits its whole family.

Theorem 1 gives a direct deferred-measurement construction for the stated pooling channel. Three consequences follow, each true of the executed model:

- 1) Coherent pooling *loses nothing* relative to measurement-based pooling for the retained-register channel; it simulates that class without mid-circuit control, with its resource cost reported explicitly below.
- 2) It does so while eliminating mid-circuit measurement, qubit reset, and classical feed-forward — capabilities that may require additional control support on a target device. The advantage asserted here is architectural and simulation-scoped rather than an empirical hardware result.
- 3) The global state remains pure from state preparation to the single terminal measurement, and the circuit is differentiable by the adjoint method end to end.

It also follows immediately that  $R_Y^{(b)}(\epsilon)$  in (14) is inert: it acts only on  $b$ , and the partial trace over  $b$  absorbs it. This is an analytic invariant of the stated retained-register channel, independent of the fixed angle's numerical value.

**Proposition 1** (Strict containment). *The family of unitary pooling channels  $\{\rho \mapsto \text{Tr}_b[V\rho V^\dagger] : V \in SU(4)\}$  strictly contains the measure-and-condition class  $\{\mathcal{M}_{\{U_m\}}\}$ .*

*Proof.* Containment is Theorem 1 applied to  $V = \sum_m |m\rangle\langle m|_b \otimes U_m$ . For strictness, take  $V = e^{i\pi/4}\text{SWAP} \in SU(4)$ . Its global phase cancels from the channel, so  $\text{Tr}_b[V\rho V^\dagger] = \rho_b$ , the full reduced state of  $b$  including its off-diagonal elements. By contrast,  $\mathcal{M}_{\{U_m\}}$  depends on  $b$  only through the diagonal blocks  $\langle m|_b \rho |m\rangle_b$ . The product states  $\rho_a \otimes |+\rangle\langle +|_b$  and  $\rho_a \otimes \frac{1}{2}I_b$  have identical  $b$ -diagonals and are therefore indistinguishable to every  $\mathcal{M}_{\{U_m\}}$ , yet  $V$  maps them to  $|+\rangle\langle +|$  and  $\frac{1}{2}I$  respectively, which are at trace distance  $\frac{1}{2}$ . □

Proposition 1 places the proposed block precisely: it realises the measurement-simulable subclass in the frozen construc-

tion, while the strict extension — non-diagonal  $V$  — represents headroom that is an empirical question rather than an architectural claim.

**Proposition 2** (Coherence-transfer functional). *Let  $\mathcal{D}_b$  be complete dephasing of  $b$  in the computational basis and define*

$$\Delta_{\text{coh}}(V) = \|\text{Tr}_b[V\rho V^\dagger] - \text{Tr}_b[V\mathcal{D}_b(\rho)V^\dagger]\|_1. \quad (21)$$

*Then  $\Delta_{\text{coh}}(W_{ab}) = 0$  for every  $\rho$ , whereas  $\Delta_{\text{coh}}(e^{i\pi/4}\text{SWAP}) > 0$ .*

*Proof.* By Theorem 1 the output of  $W_{ab}$  depends on  $\rho$  only through  $\langle m|_b \rho |m\rangle_b$ , which  $\mathcal{D}_b$  leaves invariant; hence the two terms coincide. For  $e^{i\pi/4}\text{SWAP}$  the witness in Proposition 1 gives  $\Delta_{\text{coh}} = 1$  on  $\rho_a \otimes |+\rangle\langle +|_b$  under the trace-norm definition in (21); the corresponding trace distance is  $\frac{1}{2}$ .  $\square$

2) *Numerical confirmation:* The analytic statements are also checked on the executed circuit. Evaluating the frozen  $n = 10$  model with its trained parameters on a density-matrix simulator, the coherent block and the explicit measure-and-condition channel of (16) produce terminal expectation values agreeing to a maximum deviation of  $2.2 \times 10^{-16}$  — floating-point round-off, against a tolerance of  $10^{-12}$ . Independently, inserting complete dephasing on every compressed wire immediately before pooling, at unchanged parameters, changes the readout by at most  $5.86 \times 10^{-14}$  and leaves the predicted class unchanged for all eight evaluated inputs, so the observed accuracy difference on this check is zero. The retained-wire dephasing control changes the readout by 0.4094, confirming that the null effect is specific to the compressed subsystem and its partial trace.

The retained-wire control establishes that the null result is not merely an insensitive readout: applying the same dephasing channel to retained rather than compressed wires changes the output by 0.4094, many orders of magnitude larger than the compressed-wire effect. This comparison is the canonical control archived with the dephasing evidence.

3) *Claims this does and does not support:* Theorem 1 does not support the claim that the stated coherent block preserves retained-register information that the corresponding measurement-based block destroys: for observables of this circuit, the two channels are identical. What the theorem supports instead is a narrower set of claims — exact simulation of the stated measurement-based class in the frozen construction, elimination of mid-circuit measurement, reset and feed-forward, a globally pure state throughout the pipeline, and end-to-end adjoint differentiability. Whether the strict extension of Proposition 1 yields any accuracy benefit is addressed by the pooling ablation in Section VI-C.

### G. Variational Classification Head

The classification head maps the compressed representation to a Pauli- $Z$  expectation on the first active wire. Its description follows the executed circuit rather than a simplified symbolic surrogate. On each of at most the first four active wires  $q_i$ , it applies  $R_X(\theta_{2i})$ ,  $R_Y(\theta_{2i+1})$ , and  $R_Z(\theta_{2i+8})$  in sequence.

A nearest-neighbour CNOT chain with a wraparound CNOT is followed by  $R_X(\theta_{2i+16})$  and  $R_Y(\theta_{2i+17})$  on those wires, then one additional CNOT from the first to the second active wire when at least two remain. The implementation allocates a 32-slot classifier vector and uses modular indexing so the same layout applies across supported register sizes.

A final  $R_Z(\theta_{31})$  is syntactically applied to the readout wire immediately before measuring  $Z$ . Because this gate commutes with the measured observable, it cannot change the expectation and has exactly zero gradient; it is retained only to preserve the frozen parameter layout and is not a trainable bias. For the headline ten-qubit circuit, there are 269 allocated parameter slots, 78 slots that reach the tape, and 74 effective slots at the declared  $|\partial f / \partial \theta| > 10^{-12}$  tolerance. The distinction matters because unused convolutional arrays, unpaired wires, and structurally inert gates leave allocated slots that do not influence the terminal observable.

### H. Global Parameter Scaling

Parameter sharing prevents the number of allocated slots from scaling with the number of windows, but the executed topology determines which slots are used. We therefore report allocated, tape-reaching, and effective counts separately rather than applying a closed-form count that assumes every module is independently instantiated.

#### I. Capacity accounting and executed-tape audit

The parameter vector is intentionally larger than the set of values that influence the terminal observable. Each of the four convolutional arrays is allocated as a  $4 \times 4 \times 3$  tensor, giving 48 slots per layer. Three pooling arrays reserve five three-angle pairs each, and the classifier reserves 32 slots. The headline layout therefore allocates  $4(48) + 3(5 \times 3) + 32 = 269$  slots. The executed active-wire schedule addresses only a subset of the nominal convolutional windows, and the classifier uses modular indexing; these implementation details make a single nominal parameter-count number misleading.

The 269-to-74 comparison is a parameter-allocation observation, not a generalisation theorem. Conversely, the gate count used in the capacity scaling diagnostic is 222, not 74: 48 effective slots are reused across more than one gate occurrence, with up to four occurrences driven by one slot. The two counts thus differ in opposite directions. The audit was performed both on the archived headline weights over 20 fixed regression inputs and in the independent random-initialisation gradient sweep described in Section VI-J; the agreement of the headline effective count is reported as a structural check, not as evidence of a universal effective dimension.

The same distinction matters for classical simulation. At ten qubits, the statevector has  $2^{10} = 1024$  complex amplitudes and is directly tractable on a classical computer. The small state space and shallow executed tape are the reasons every result in this paper can be checked exactly or with explicitly declared finite-shot local simulation. They are not evidence of a quantum computational advantage. In particular, the resource counts below separate state preparation from the model body

TABLE III

HEADLINE CAPACITY AND OPERATION ACCOUNTING. THE DISTINCT ROWS ANSWER DIFFERENT QUESTIONS AND MUST NOT BE SUBSTITUTED FOR ONE ANOTHER.

| Quantity                        | Value | Definition                                        |
|---------------------------------|-------|---------------------------------------------------|
| Allocated parameter slots       | 269   | Reserved by the model layout                      |
| Tape-reaching slots             | 78    | Slots syntactically read by the frozen tape       |
| Effective slots                 | 74    | $ \partial f / \partial \theta  > 10^{-12}$ audit |
| Trainable gates on tape         | 222   | Parameterised gate occurrences                    |
| Gates driven by effective slots | 218   | Occurrences from the 74 live slots                |
| Total tape operations           | 327   | Trainable and fixed operations                    |

so that the cost of loading a classical image is not hidden behind the variational circuit.

### J. Architectural Interpretation

The FQCNN implements global amplitude state preparation, shared register-index convolutional mixing, coherent dimensional compression, and variational classification before a terminal Pauli- $Z$  expectation. Its effective capacity is described by the executed-tape and gradient audits above; no expressive benefit is attributed to the inert terminal  $R_Z$ .

## V. TRAINING AND OPTIMIZATION

### A. Learning Objective and Risk Functional

The learning objective is governed by empirical risk minimisation. The model maps the quantum expectation value  $f_\Theta(x^{(k)})$  to a bounded probability and maps labels to  $\tilde{y}^{(k)} \in \{0, 1\}$ :

$$p^{(k)} = \frac{\text{clip}(f_\Theta(x^{(k)}), -1, 1) + 1}{2}. \quad (22)$$

The executed class-weighted Binary Cross-Entropy (BCE) objective is

$$\mathcal{L}(\Theta) = -\frac{1}{M} \sum_{k=1}^M \left[ w_+ \tilde{y}^{(k)} \log(p^{(k)} + \varepsilon) + w_- (1 - \tilde{y}^{(k)}) \log(1 - p^{(k)} + \varepsilon) \right], \quad (23)$$

where  $M$  is the mini-batch size,  $w_+$  and  $w_-$  are computed from the training split only, and  $\varepsilon = 10^{-7}$  prevents singular logarithms. This is a weighted Bernoulli negative log-likelihood.

### B. Measurement Estimation Model

On hardware the expectation value is estimated through repeated measurements. With  $S$  shots, the estimator is

$$\hat{f}_\theta(x) = \frac{1}{S} \sum_{n=1}^S z^{(n)}, \quad z^{(n)} \in \{-1, +1\}. \quad (24)$$

It satisfies  $\mathbb{E}[\hat{f}_\theta] = f_\theta$  and  $\text{Var}[\hat{f}_\theta] = (1 - f_\theta^2)/S$ , so the estimator is unbiased and its variance decreases as  $\mathcal{O}(1/S)$ .

### C. Gradient Evaluation

The reported statevector training uses the simulator's adjoint/backpropagation differentiation path, with the sequential `lightning.qubit` execution retained as the correctness oracle. For a hardware implementation, gates with a two-eigenvalue generator admit an exact occurrence-level parameter-shift estimator; this is included only as a cost model and is not the differentiation method used for the reported results. In the interior of the clipping interval, the executed BCE formulation gives

$$\frac{\partial \mathcal{L}}{\partial \theta} = -\frac{1}{2M} \sum_{k=1}^M \left[ \frac{w_+ \tilde{y}^{(k)}}{p^{(k)} + \varepsilon} - \frac{w_- (1 - \tilde{y}^{(k)})}{1 - p^{(k)} + \varepsilon} \right] \frac{\partial f_\Theta(x^{(k)})}{\partial \theta}. \quad (25)$$

where, for one eligible gate occurrence  $j$ ,  $\partial f / \partial \theta_j = \frac{1}{2}(f_{j,+} - f_{j,-})$  with shifts  $\pm\pi/2$ . If one trainable slot is reused in  $r$  gate occurrences, the chain rule sums those occurrence-level derivatives, requiring up to  $2r$  shifted circuit evaluations absent a specialised grouped rule. This is a hardware cost model, not the differentiation method used for the reported simulator results.

### D. Optimization Procedure

The optimisation loop utilises the Adam optimiser to adapt step sizes throughout the parameter landscape. The historical headline run used  $\eta = 0.005$ , mini-batches of  $M = 32$ , gradient clipping, and an Exponential Moving Average (EMA) with decay  $\gamma = 0.99$  for at most 50 epochs with early stopping. The bounded Q1 comparison uses its separately declared 30-epoch protocol. These settings are reported as finite experimental choices, not as a proof that barren plateaus are absent.

### E. Reproducible training protocol

The comparison campaign treats data identity, split identity, model selection, and test evaluation as separate objects. The protocol is summarised in Algorithm 1. In particular, a run cannot become a valid comparison cell merely because it produced a metric: its configuration, split identifier, ordered test sample identifiers, completion state, weights, and prediction archive must all agree with the registered task.

The 666-example pool is a protocol consequence rather than a hidden sample count: rounding the fixed fractions

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**Algorithm 1** Registered training and evaluation protocol

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- 1: Select a dataset family and binary class pair from the frozen task list.
  - 2: Load the checksum-validated source snapshot and retain source-stable IDs.
  - 3: Form the 666-example task pool and create a deterministic stratified 60/15/25 manifest.
  - 4: Apply the fixed preprocessing and amplitude encoding; preserve the manifest ID.
  - 5: **for** each arm and seed in the declared matrix **do**
  - 6:   Initialise the arm with the recorded seed and frozen configuration.
  - 7:   Train on the training partition only; monitor validation loss.
  - 8:   Restore the validation-selected checkpoint and record its selection metadata.
  - 9:   Evaluate the test partition once and save ordered IDs, labels, outputs, and metrics.
  - 10: **end for**
  - 11: Aggregate only complete cells with matching manifests and configuration hashes.
  - 12: Compute seed-level uncertainty and paired tests from the retained archives.
- 

gives 400 training, 100 validation, and 166 test examples. The sample pool is selected before any test evaluation and the same ordered identifiers are supplied to every arm. A test label is never used for checkpoint selection, early stopping, or hyperparameter choice. The historical 12,665-image MNIST run follows a different, larger split and is retained only as provenance; it is not merged into the Q1 aggregate.

The main simulator path uses PennyLane’s `default.qubit` backpropagation device for batched training and retains `lightning.qubit` as a sequential correctness oracle. A separate regression suite verifies agreement of outputs, losses, all 269 parameter gradients, and one Adam update between the two paths. This engineering check is relevant to reproducibility: the faster training path is not a second circuit definition. It calls the same canonical circuit builder and is subject to the same frozen topology signature.

The local preprocessing function min–max normalises the image matrix to  $[0, 1]$ , flattens each native  $28 \times 28$  image, pads it with zeros to 1024 entries, and maps the two source labels to  $-1$  and  $+1$ . The amplitude embedding then performs the per-state  $\ell_2$  normalisation required by state preparation. The preprocessing mode, image size, qubit count, encoding type, pooling mode, seed, and split ID are all written to run metadata. This is a fixed-input pipeline choice, not a learned feature-extraction stage, and its global nature is why the manuscript does not attribute image-space locality to the model.

## F. Computational Complexity

For the simulator path, adjoint/backpropagation cost is determined by the retained state and circuit evaluation rather than by parameter-shift executions. If parameter-shift is used on hardware, its leading evaluation count is two shifted circuits per eligible gate occurrence and datapoint, with additional accounting for reused slots and shot cost set by the measurement budget. Statevector simulation scales as  $\mathcal{O}(2^n)$  in memory. The resource and any local-noise evidence reported below are explicitly simulation/fake-backend evidence; they do not establish hardware compatibility or deployment readiness.

## G. Noise and Gradient Stability

1) *Statistical measurement noise*: With  $\rho(\Theta; x) = |\psi(\Theta; x)\rangle\langle\psi(\Theta; x)|$  and Born’s rule,  $f_\theta(x) = \text{Tr}[\rho Z_0]$ . Each shifted expectation estimator is unbiased with variance scaling as  $\mathcal{O}(1/S)$ , so their linear parameter-shift estimate of  $\partial f/\partial\theta$  is unbiased. A finite-shot plug-in estimator of the Binary Cross-Entropy gradient need not be unbiased, however, because the loss is nonlinear in the estimated probability and requires clipping near 0 and 1.

2) *Hardware-induced quantum noise*: Hardware noise replaces unitary evolution with a completely positive trace-preserving map  $\rho_{\text{noisy}} = \varepsilon(\rho)$ . Under the stated global depolarising model  $\varepsilon_{\text{dep}}(\rho) = (1-p)\rho + pI/2^n$ , a traceless Pauli- $Z$  expectation is attenuated by  $(1-p)$ . Under the phase-flip model  $\varepsilon_{\text{phase}}(\rho) = (1-p)\rho + pZ\rho Z$ , computational-basis off-diagonal terms are rescaled according to the affected wire. Neither identity alone predicts end-to-end accuracy; the bounded local simulations below evaluate specified channels and noise levels.

3) *Noise propagation*: Entangling operations have bounded support in the register-index topology, but global amplitude encoding prevents this fact from being interpreted as a bound on image-space noise propagation. End-to-end noisy accuracy is treated as an empirical question and is reported only for the declared local validation protocol.

4) *Effect on gradients*: If the noise channel is independent of  $\theta$ , differentiation moves inside the trace,  $\partial f/\partial\theta = \text{Tr}[\varepsilon(\partial\rho/\partial\theta)Z_0]$ . The resulting gradient behavior depends on the channel and circuit; no universal direction-preservation or trainability guarantee is inferred here.

## VI. EXPERIMENTAL RESULTS AND ANALYSIS

### A. Dataset Description

The original headline experiment used the MNIST handwritten-digit dataset [15] in the standard IDX binary format, selecting digits 0 and 1 for a strict binary task. Its single seeded stratified 60/15/25 split and 12,665-image count are retained as historical provenance only and are summarised in Table V. The Q1 evidence uses the frozen multi-domain registry and records source checksums, ordered sample identifiers, split hashes, validation-only model selection, and exactly one final test evaluation for every seed. Table VI gives the registered task definitions and split sizes; the complete source-file hashes and sample manifests remain

TABLE IV

PROTOCOL CONTROLS USED TO MAKE THE WITHIN-STUDY COMPARISON AUDITABLE. THESE CONTROLS DESCRIBE THE UNIT OF INFERENCE; THEY DO NOT TURN THE FINITE CAMPAIGN INTO AN ESTIMATE OF ALL POSSIBLE DATASETS OR TRAINING PROCEDURES.

| Control             | Implementation                                                                                                                 | Reason for reporting it                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Task registration   | Four binary tasks are listed before aggregation: MNIST 0-vs-1, MNIST 3-vs-5, Fashion-MNIST 0-vs-6, and KMNIST 2-vs-3.          | Separates continuity, harder in-domain behavior, and two dataset-family checks from post hoc task selection.       |
| Split identity      | Each seed has a persisted stratified manifest with source-stable sample IDs and a SHA-256-derived split ID.                    | Prevents an arm from silently using a different partition or a reconstructed random split.                         |
| Model selection     | Validation loss controls checkpoint restoration and early stopping; the test partition is read once at the end.                | Prevents test-driven selection and keeps the final metric attached to the selected checkpoint.                     |
| Capacity disclosure | FQCNN allocation, tape usage, effective slots, logistic coefficients, and MLP parameters are reported separately.              | Avoids calling the two-unit MLP parameter-matched when its dense parameter count is larger.                        |
| Seed aggregation    | Five registered seeds are paired by task and split; population standard deviation and seeded bootstrap intervals are reported. | Displays training variability and avoids treating individual test examples as independent repetitions of training. |
| Artifact validation | Complete status, expected configuration, weights, predictions, IDs, and metric schema are checked before aggregation.          | Excludes partial, stale, wrong-split, or malformed outputs from the scientific tables.                             |

TABLE V  
HISTORICAL HEADLINE DATASET CONFIGURATION

| Property           | Value                                               |
|--------------------|-----------------------------------------------------|
| Dataset            | MNIST (LeCun et al.)                                |
| Format             | IDX binary (ubyte)                                  |
| Binary classes     | Digit 0 (−1) vs Digit 1 (+1)                        |
| Total samples      | 12,665 (5,923 digit 0, 6,742 digit 1)               |
| Training samples   | 7,599 (60%, stratified)                             |
| Validation samples | 1,900 (15%, stratified)                             |
| Test samples       | 3,166 (25%, stratified)                             |
| Input image size   | $28 \times 28$ grayscale                            |
| Normalization      | Min-max scaling to $[0, 1]$                         |
| Encoding strategy  | Global amplitude encoding                           |
| State amplitudes   | 1024 ( $2^{10}$ , zero-padded)                      |
| Label mapping      | Digit 0 $\rightarrow -1$ , Digit 1 $\rightarrow +1$ |

in the machine-readable provenance artifact. The Fashion-MNIST and KMNIST snapshots are identified by upstream URLs and checksums in that artifact rather than by non-journal dataset preprints.

Each input is min-max normalised to  $[0, 1]$ , flattened to 784 values, zero-padded to 1024, and embedded directly into one ten-qubit state by global amplitude encoding. This state preparation does not preserve a spatial-neighbour relation between image pixels and register indices; all locality claims in the variational circuit are therefore register-index statements. Labels are mapped to  $\{+1, -1\}$  for the Pauli- $Z$  expectation output. The corrected Q1 campaign uses 666 source examples per task and the frozen 60/15/25 split, yielding exactly 400 training, 100 validation, and 166 test examples for each seed; every arm consumes the same ordered identifiers. The historical 0-vs-1 task is a calibration reference, while the Q1 matrix adds a harder MNIST pair and two independently checksummed dataset families.

### B. Experimental Setup and Training Configuration

The experimental validation is executed using numerical quantum simulation backends. The architecture uses a register of  $n = 10$  qubits initialised in the ground state  $|0\rangle^{\otimes 10}$ . A 2D input image  $X \in \mathbb{R}^{28 \times 28}$  is flattened into a 784-dimensional vector  $x$ , which is zero-padded to 1024 and mapped to the 1024 state amplitudes ( $2^{10}$ ) using a global amplitude-encoding sub-circuit. Simulations are run via the `lightning.qubit` exact statevector simulator. The corrected Q1 campaign uses 666 source examples per task and the frozen 60/15/25 split, yielding exactly 400 training, 100 validation, and 166 test examples; the historical 12,665-sample count belongs only to the single-seed headline provenance described above. The model comprises convolutional layers with shared  $SU(2)$  kernels, coherence-preserving pooling layers, and a final variational classification head. Training uses mini-batch optimisation with Adam, gradient clipping, and EMA of parameters. The exact Q1 settings are listed in Table VII.

### C. Results

The previously reported optimisation profile used a single seeded MNIST reference run with a 0.005 learning rate and a 50-epoch maximum; it is retained as provenance, not as the Q1 headline result. The final performance statement is taken from the validated five-seed, four-task aggregate and reports uncertainty and paired baseline tests; the old 3,166-image confusion counts are not reused as multi-domain evidence.

For each task and arm, accuracy is summarised across the five seed-level test evaluations by its mean and population standard deviation. The primary paired test uses the five per-seed accuracies in a two-sided Wilcoxon signed-rank test; the twelve task-comparator tests are corrected as one Holm-Bonferroni family at  $\alpha = 0.05$ . The reported 95% intervals are seeded percentile bootstrap intervals for the mean using 10,000

TABLE VI

REGISTERED Q1 EVALUATION TASKS AND PROVENANCE. EACH ROW USES 666 SOURCE EXAMPLES AND THE FROZEN 400/100/166 TRAIN/VALIDATION/TEST SPLIT FOR SEEDS 0–4; SOURCE-FILE CHECKSUMS AND ORDERED IDENTIFIERS ARE RECORDED IN THE CITED ARTIFACT.

| Task           | Dataset/classes                    | Role                   | Split       | Seeds | Evidence               |
|----------------|------------------------------------|------------------------|-------------|-------|------------------------|
| MNIST 0-vs-1   | MNIST (0, 1)                       | continuity/calibration | 400/100/166 | 0–4   | provenance + aggregate |
| MNIST 3-vs-5   | MNIST (3, 5)                       | harder in-domain       | 400/100/166 | 0–4   | provenance + aggregate |
| Fashion 0-vs-6 | Fashion-MNIST (T-shirt/top, Shirt) | cross-domain clothing  | 400/100/166 | 0–4   | provenance + aggregate |
| KMNIST 2-vs-3  | Kuzushiji-MNIST (2, 3)             | distinct script domain | 400/100/166 | 0–4   | provenance + aggregate |

TABLE VII

TRAINING CONFIGURATION FOR THE VALIDATED Q1 COMPARISON.

| Parameter      | Value                                       |
|----------------|---------------------------------------------|
| Epochs         | 30 maximum; patience 3                      |
| Batch size     | 32                                          |
| Learning rate  | 0.02                                        |
| Optimizer      | Adam                                        |
| Loss function  | Weighted Binary Cross-Entropy               |
| Quantum device | lightning.qubit (PennyLane)                 |
| Shots          | None (exact statevector)                    |
| Seeds          | 0, 1, 2, 3, 4                               |
| Per-task split | 400 / 100 / 166 (train / validation / test) |

TABLE VIII

WITHIN-STUDY COMPARISON PROTOCOL. VALUES ARE GENERATED FROM THE VALIDATED Q1 AGGREGATE; HETEROGENEOUS LITERATURE PERCENTAGES ARE NOT RANKED AGAINST THESE RESULTS. TTN DENOTES THE BALANCED TREE TENSOR NETWORK OF GRANT ET AL. [13].

| Method           | Pool  | Coh.  | Params | Source       |
|------------------|-------|-------|--------|--------------|
| FQCNN (ours)     | U     | Yes   | 269/74 | Q1 aggregate |
| Logistic         | –     | –     | 785    | Q1 aggregate |
| MLP (2-unit)     | –     | –     | 1,573  | Q1 aggregate |
| TTN              | U     | Yes   | 48     | Q1 aggregate |
| Prior literature | Mixed | Mixed | –      | Context only |

resamples. Exact McNemar tests are retained as descriptive per-seed diagnostics on paired correctness vectors; no pooled McNemar p-value is used for confirmatory inference. With five paired seeds, these tests have limited power, so non-significance is not interpreted as equivalence.

#### D. Analysis of Model Performance

The unitary construction preserves the global state until the terminal observable, and the pooling theorem establishes exact equivalence to the corresponding measure-and-condition channel for the stated block. The empirical contribution is assessed by the Q1 aggregate, pooling-transfer ablation, finite-size diagnostics, and local resource/noise evidence; no accuracy advantage, universal trainability, or generalisation guarantee is inferred from the historical single-seed gap.

#### E. Comparison with Existing QCNN Approaches

The primary comparison is within-study: logistic regression, a compact two-unit dense MLP, and the balanced tree tensor network (TTN) of Grant et al. [13] are evaluated on the same ordered samples, split manifests, and seed set as the FQCNN. The arm definitions and parameter accounting are given in

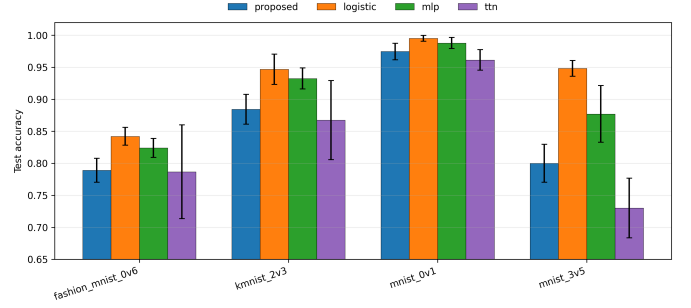


Fig. 4. Q1 comparison accuracy (mean  $\pm$  standard deviation over five seeds) generated directly from Results/evidence/q1\_comparison.json. The figure is a descriptive within-study comparison; it is not a quantum-advantage claim.

Table VIII. The logistic model has 785 trainable coefficients for the 784 flattened pixels, the two-unit MLP has 1,573 dense parameters, and the ten-qubit TTN has 48 trainable parameters under the executed consecutive-pair tree schedule. The MLP is therefore not parameter-matched to the FQCNN’s 269 allocated/74 effective quantum slots, and this conservative capacity mismatch is disclosed rather than hidden. Table IX and Fig. 4 show the accuracy result, while Table X reports the FQCNN metric vector. The aggregate does not show FQCNN dominance: it is below logistic regression and the MLP on all four tasks, while exceeding TTN on all four in mean accuracy (with uncertainty and multiplicity-adjusted tests reported in the machine-readable artifact). We retain this bounded result rather than converting the architectural contribution into an accuracy claim. Prior literature uses different datasets, preprocessing, parameter budgets, and evaluation protocols, so its percentages are context rather than a ranked baseline. The pooling theorem separates the architectural distinction (removing mid-circuit control) from any empirical accuracy question; relative-performance statements are restricted to the aggregate’s paired effects and multiplicity-adjusted tests.

1) *Cross-task interpretation:* The per-seed values in Table XI show why the seed distribution is more informative than a single headline score. On MNIST 0-vs-1, every arm is strong and the FQCNN reaches  $0.975 \pm 0.013$ , so the task mainly verifies continuity with the historical implementation. The harder MNIST 3-vs-5 task separates the hierarchical quantum arm from the linear baseline: FQCNN accuracy is  $0.800 \pm 0.030$ , TTN is  $0.730 \pm 0.047$ , and the logistic baseline is  $0.948 \pm 0.012$ . This is a useful positive result for the

TABLE IX

VALIDATED Q1 COMPARISON ACCURACY (MEAN  $\pm$  STANDARD DEVIATION OVER FIVE SEEDS). ALL ARMS USE THE SAME ORDERED SAMPLES, SPLIT MANIFEST, AND SINGLE FINAL TEST EVALUATION; COMPLETE METRICS AND PAIRED TESTS ARE ARCHIVED IN `RESULTS/EVIDENCE/Q1_COMPARISON.JSON`.

| Task              | FQCNN             | Logistic          | MLP               | TTN               |
|-------------------|-------------------|-------------------|-------------------|-------------------|
| fashion_mnist_0v6 | 0.789 $\pm$ 0.019 | 0.842 $\pm$ 0.014 | 0.824 $\pm$ 0.015 | 0.787 $\pm$ 0.073 |
| kmnist_2v3        | 0.884 $\pm$ 0.023 | 0.947 $\pm$ 0.024 | 0.933 $\pm$ 0.016 | 0.867 $\pm$ 0.062 |
| mnist_0v1         | 0.975 $\pm$ 0.013 | 0.995 $\pm$ 0.005 | 0.988 $\pm$ 0.009 | 0.961 $\pm$ 0.016 |
| mnist_3v5         | 0.800 $\pm$ 0.030 | 0.948 $\pm$ 0.012 | 0.877 $\pm$ 0.044 | 0.730 $\pm$ 0.047 |

TABLE X

VALIDATED Q1 METRICS FOR FQCNN (MEAN  $\pm$  POPULATION STANDARD DEVIATION OVER FIVE SEEDS). THESE VALUES EXPOSE THE METRIC-LEVEL RESULT BEHIND THE ACCURACY COMPARISON; THE CORRESPONDING BASELINE METRICS, CONFIDENCE INTERVALS, AND PAIRED EFFECTS ARE RETAINED IN THE AGGREGATE ARTIFACT.

| Task              | Accuracy          | Balanced acc.     | F1                | ROC-AUC           | PR-AUC            |
|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| fashion_mnist_0v6 | 0.789 $\pm$ 0.019 | 0.789 $\pm$ 0.019 | 0.784 $\pm$ 0.021 | 0.863 $\pm$ 0.017 | 0.842 $\pm$ 0.045 |
| kmnist_2v3        | 0.884 $\pm$ 0.023 | 0.884 $\pm$ 0.023 | 0.893 $\pm$ 0.020 | 0.968 $\pm$ 0.005 | 0.969 $\pm$ 0.003 |
| mnist_0v1         | 0.975 $\pm$ 0.013 | 0.975 $\pm$ 0.013 | 0.975 $\pm$ 0.013 | 0.997 $\pm$ 0.002 | 0.998 $\pm$ 0.002 |
| mnist_3v5         | 0.800 $\pm$ 0.030 | 0.800 $\pm$ 0.030 | 0.776 $\pm$ 0.033 | 0.901 $\pm$ 0.035 | 0.896 $\pm$ 0.040 |

TABLE XI

PER-SEED ACCURACY VALUES UNDERLYING THE PRIMARY COMPARISON. THE SEED ORDER IS 0, 1, 2, 3, 4 FOR EVERY COLUMN; VALUES ARE SHOWN TO FOUR DECIMAL PLACES TO MAKE THE SEED-LEVEL PAIRING VISIBLE.

| Task           | Arm      | Seed 0 | Seed 1 | Seed 2 | Seed 3 | Seed 4 |
|----------------|----------|--------|--------|--------|--------|--------|
| Fashion 0-vs-6 | FQCNN    | 0.8072 | 0.7952 | 0.8072 | 0.7771 | 0.7590 |
|                | Logistic | 0.8313 | 0.8614 | 0.8373 | 0.8554 | 0.8253 |
|                | MLP      | 0.8072 | 0.8434 | 0.8253 | 0.8373 | 0.8072 |
|                | TTN      | 0.8072 | 0.8012 | 0.8313 | 0.8494 | 0.6446 |
| KMNIST 2-vs-3  | FQCNN    | 0.8855 | 0.8554 | 0.9036 | 0.9157 | 0.8614 |
|                | Logistic | 0.9819 | 0.9096 | 0.9578 | 0.9458 | 0.9398 |
|                | MLP      | 0.9518 | 0.9157 | 0.9518 | 0.9277 | 0.9157 |
|                | TTN      | 0.9277 | 0.7892 | 0.9096 | 0.9157 | 0.7952 |
| MNIST 0-vs-1   | FQCNN    | 0.9759 | 0.9639 | 0.9578 | 0.9819 | 0.9940 |
|                | Logistic | 1.0000 | 0.9880 | 0.9940 | 0.9940 | 1.0000 |
|                | MLP      | 0.9940 | 0.9819 | 0.9880 | 0.9759 | 1.0000 |
|                | TTN      | 0.9819 | 0.9578 | 0.9699 | 0.9337 | 0.9639 |
| MNIST 3-vs-5   | FQCNN    | 0.7651 | 0.8133 | 0.7771 | 0.7952 | 0.8494 |
|                | Logistic | 0.9398 | 0.9337 | 0.9518 | 0.9458 | 0.9699 |
|                | MLP      | 0.8675 | 0.8795 | 0.9036 | 0.8012 | 0.9337 |
|                | TTN      | 0.7651 | 0.7470 | 0.6386 | 0.7590 | 0.7410 |

TABLE XII

SEED-LEVEL PAIRED EFFECTS FOR THE PRIMARY COMPARISON. EACH ENTRY IS FQCNN ACCURACY MINUS COMPARATOR ACCURACY, FOLLOWED BY THE UNADJUSTED AND HOLM-ADJUSTED TWO-SIDED WILCOXON  $p$  VALUES. THE TWELVE TESTS FORM ONE DECLARED FAMILY.

| Task                 | FQCNN – Logistic                     | FQCNN – MLP                          | FQCNN – TTN                           |
|----------------------|--------------------------------------|--------------------------------------|---------------------------------------|
| Fashion-MNIST 0-vs-6 | −0.0530; $p = .0625$ , $p_H = .7500$ | −0.0349; $p = .0656$ , $p_H = .7500$ | +0.0024; $p = .7150$ , $p_H = 1.0000$ |
| KMNIST 2-vs-3        | −0.0627; $p = .0625$ , $p_H = .7500$ | −0.0482; $p = .0625$ , $p_H = .7500$ | +0.0169; $p = .4652$ , $p_H = 1.0000$ |
| MNIST 0-vs-1         | −0.0205; $p = .0625$ , $p_H = .7500$ | −0.0133; $p = .1250$ , $p_H = .7500$ | +0.0133; $p = .6250$ , $p_H = 1.0000$ |
| MNIST 3-vs-5         | −0.1482; $p = .0625$ , $p_H = .7500$ | −0.0771; $p = .0625$ , $p_H = .7500$ | +0.0699; $p = .0679$ , $p_H = .7500$  |

comparison with TTN, but it is not evidence that the quantum model outperforms a strong classical classifier.

The cross-domain tasks expose a different pattern. FQCNN reaches  $0.789 \pm 0.019$  on Fashion-MNIST 0-vs-6 and  $0.884 \pm 0.023$  on KMNIST 2-vs-3, while logistic regression remains

higher on both. The TTN variance is notably larger on Fashion-MNIST (0.073 versus 0.019 for FQCNN), whereas the mean accuracies are nearly tied. On KMNIST, FQCNN exceeds TTN by roughly two percentage points in mean accuracy, again without a multiplicity-adjusted significant paired

TABLE XIII

POOLING-TRANSFER ABLATION ON FASHION-MNIST 0-VS-6 (THREE SEEDS). HOLM-ADJUSTED  $p$  VALUES ARE TWO-SIDED SEED-LEVEL WILCOXON TESTS VERSUS THE UNITARY REFERENCE OVER FOUR VARIANTS.

| Pooling variant     | Accuracy          | $p_{\text{Holm}}$ |
|---------------------|-------------------|-------------------|
| Unitary (reference) | $0.803 \pm 0.006$ | —                 |
| No pooling          | $0.815 \pm 0.006$ | 0.719             |
| Measurement         | $0.803 \pm 0.006$ | 1.000             |
| Coherent            | $0.791 \pm 0.023$ | 1.000             |
| SU(4)               | $0.801 \pm 0.013$ | 1.000             |

effect. These patterns support a more precise conclusion: the architecture is a reproducible hierarchical quantum comparator whose finite behavior depends strongly on the task, not a uniformly competitive replacement for classical image baselines.

Table XII makes the inferential limitation explicit. The FQCNN–logistic and FQCNN–MLP effects are negative for every task, and the FQCNN–TTN effects are positive for every task. Nevertheless, the five-seed Wilcoxon tests do not survive the declared Holm correction. This combination of consistent direction and limited power is reported as an observation, not as proof of equivalence or superiority. The test unit is the seed-level score; the 166 test examples within a split do not create additional independent training replicates.

The transfer experiment in Table XIII uses the same 166-example test set and clean checkpoints for seeds 0–2. It is a finite ablation rather than a universal ranking: no variant differs from the unitary reference after Holm correction, and the small three-seed sample is not powered to establish equivalence. Per-seed McNemar values are retained as descriptive diagnostics only.

#### F. Secondary pooling-family evidence

The completed E3 pooling-arm campaign provides a complementary, smaller-budget view of the pooling family. It used the headline ten-qubit geometry, 400 training examples, 30 epochs, five seeds, and three binary tasks, with 142 test examples per task and split. These runs are reported separately from the primary Q1 matrix because they predate its four-task, 666-example registration; they are not used to replace the primary aggregate or to select a new headline architecture.

Across these secondary tasks, removing pooling lowers the aggregate mean by 2.7 percentage points relative to the frozen unitary arm and is significant in the archived paired tests. The measurement and unitary rows tie exactly in the statevector implementation, as expected from the theorem and deferred measurement. The coherent extension improves the aggregate by only 0.19 percentage points and is not significant. The SU(4) arm has a 2.4 percentage point aggregate advantage, with the largest task-level separation on MNIST 3-vs-5; the seed-level Wilcoxon result does not confirm that advantage after the declared correction. This is family headroom, not evidence that the frozen three-angle block is optimal, and it is why the frozen arm remains the reference construction throughout this paper.

#### G. Information dynamics across the pooling cascade

The word “coherent” can refer either to the purity of the complete state or to the coherence of a reduced subsystem. These are not interchangeable. To make the distinction observable, we recorded reduced-state diagnostics at the encoded state, immediately before and after each pooling stage, and after the classifier for eight archived-weight inputs. For a reduced density matrix  $\rho_A$ , the reported  $\ell_1$  coherence is

$$C_{\ell_1}(\rho_A) = \sum_{i \neq j} |(\rho_A)_{ij}|, \quad (26)$$

and the entropy and mutual information are

$$S(\rho_A) = -\text{Tr}(\rho_A \log_2 \rho_A), \quad I(A : B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB}). \quad (27)$$

Purity is  $\text{Tr}(\rho_A^2)$ . Entropies and mutual information in Table XV are in bits; entries are mean  $\pm$  standard deviation over the eight inputs.

The encoded ten-wire state has unit purity and near-zero entropy, as expected for a pure state preparation. After tracing out the compressed wires, the active reduced state is mixed because it is entangled with the inactive subsystem; this is compatible with a globally pure unitary computation. By the final readout, the one-wire reduced state has purity approximately 0.588 and entropy approximately 0.869 bits, so “fully coherent” should not be read as “locally pure” or “classically inaccessible.”

The mutual information falls from approximately 7.814 bits before the first pool to 0.204 bits before the second and 0.005 bits before the third. The last pooling stage therefore has little measured keep–discard correlation on these archived inputs. Pooling changes the entropy by less than 0.07 bits at each boundary, and the final classifier changes basis-dependent coherence without changing the reduced state’s spectrum. These observations qualify the architectural language: the controlled rotations provide a prescribed gate path for consolidation, but the diagnostic does not demonstrate that each pool concentrates useful information or that the deepest stage is active for every task.

The information audit also supplies a control for the partial-trace description. If the reported changes were merely an artifact of an insensitive readout, a local operation on the retained wires would be expected to have the same null effect. Instead, dephasing the retained wires changes the readout by 0.4094 on the fixed-input control, whereas dephasing the compressed wires changes it by at most  $5.86 \times 10^{-14}$ . The difference is consistent with the theorem’s specific subsystem statement and does not imply that the discarded subsystem is irrelevant to the global state.

#### H. Noise Robustness

The pooling theorem and dephasing controls establish circuit-level identities, not end-to-end noise robustness. The Q1 package therefore reports a bounded local validation experiment using clean checkpoints, declared depolarising/dephasing channels, fixed validation subsets, and an isolated Qiskit/Aer environment. The protocol covers MNIST

TABLE XIV

SECONDARY E3 POOLING-ARM COMPARISON. ENTRIES ARE TEST ACCURACY MEANS  $\pm$  POPULATION STANDARD DEVIATION OVER FIVE SEEDS FOR EACH TASK. EACH ARM HAS 15 CELLS IN TOTAL (THREE TASKS AND FIVE SEEDS); THESE MATCHED-BUDGET RUNS ARE DESCRIPTIVE SUPPORT FOR THE POOLING INTERPRETATION, NOT THE PRIMARY Q1 COMPARISON.

| Task         | No pooling          | Measurement         | Unitary             | Coherent            | SU(4)               |
|--------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| MNIST 0-vs-1 | $0.9507 \pm 0.0134$ | $0.9662 \pm 0.0082$ | $0.9662 \pm 0.0082$ | $0.9634 \pm 0.0113$ | $0.9718 \pm 0.0161$ |
| MNIST 3-vs-5 | $0.7620 \pm 0.0376$ | $0.8070 \pm 0.0440$ | $0.8070 \pm 0.0440$ | $0.8099 \pm 0.0134$ | $0.8634 \pm 0.0451$ |
| MNIST 4-vs-9 | $0.8394 \pm 0.0113$ | $0.8606 \pm 0.0233$ | $0.8606 \pm 0.0233$ | $0.8662 \pm 0.0461$ | $0.8704 \pm 0.0580$ |

TABLE XV

PER-STAGE INFORMATION DIAGNOSTICS FOR THE FROZEN TEN-QUBIT CIRCUIT. THE REDUCED STATE IS TAKEN ON THE ACTIVE/KEPT WIRES; MUTUAL INFORMATION IS REPORTED ONLY AT POOLING BOUNDARIES. VALUES ARE FROM THE ARCHIVED-WEIGHT E4 ARTIFACT AND ARE DESCRIPTIVE, NOT CAUSAL ATTRIBUTION OF ACCURACY.

| Stage            | Kept wires | $C_{\ell_1}$        | Purity              | $S$                 | $I(\text{keep} : \text{discard})$ |
|------------------|------------|---------------------|---------------------|---------------------|-----------------------------------|
| Encoded          | 10         | $767.262 \pm 7.232$ | $1.0000 \pm 0.0000$ | $0.0000 \pm 0.0000$ | —                                 |
| Before pooling 0 | 5          | $5.906 \pm 0.206$   | $0.0919 \pm 0.0037$ | $3.9070 \pm 0.0321$ | $7.814 \pm 0.064$                 |
| After pooling 0  | 5          | $6.567 \pm 0.170$   | $0.0942 \pm 0.0030$ | $3.8901 \pm 0.0309$ | $7.780 \pm 0.062$                 |
| Before pooling 1 | 2          | $0.819 \pm 0.051$   | $0.3356 \pm 0.0051$ | $1.7565 \pm 0.0120$ | $0.204 \pm 0.009$                 |
| After pooling 1  | 2          | $0.697 \pm 0.038$   | $0.3134 \pm 0.0044$ | $1.8214 \pm 0.0111$ | $0.358 \pm 0.013$                 |
| Before pooling 2 | 1          | $0.403 \pm 0.017$   | $0.5923 \pm 0.0072$ | $0.8624 \pm 0.0111$ | $0.005 \pm 0.002$                 |
| After pooling 2  | 1          | $0.332 \pm 0.022$   | $0.5879 \pm 0.0074$ | $0.8691 \pm 0.0114$ | $0.016 \pm 0.004$                 |
| After classifier | 1          | $0.253 \pm 0.016$   | $0.5879 \pm 0.0074$ | $0.8691 \pm 0.0114$ | —                                 |

3-vs-5 and Fashion-MNIST 0-vs-6, seeds 0–2, 25 validation samples per seed, noise levels  $p \in \{0, 0.005, 0.01, 0.02\}$ , and 128 shots at nonzero levels; checkpoints are not retrained and the test split is never used. The zero-noise agreement check passes, while the finite-shot nonzero results vary across task, seed, channel, and level without a monotone pattern. Those values are simulator evidence only; no accuracy-retention or device-noise claim is made beyond this declared scope.

The exact statevector and Aer statevector outputs agree at zero noise to a maximum absolute difference below  $1.34 \times 10^{-15}$  across the six checkpoint records. This establishes that the local noise harness begins from the canonical circuit rather than from a silently modified model. It does not validate the noisy channel against a physical device. The finite-shot columns are intentionally presented as a sensitivity probe: with only 25 validation examples per checkpoint, a change in a few signs can move the accuracy by a large amount, and increasing  $p$  need not produce a monotone sequence when the checkpoint is fixed and shot noise is present.

### I. Scalability and Practicality

Amplitude encoding fixes the state-vector width at the declared qubit count, while parameter sharing controls the number of variational slots. Table XVII reports total logical, decomposed, and transpiled depths for  $n = 4, 6, 8, 10$ , including state preparation and readout. The transpilation target is the local Qiskit “FakePeekskill” BackendV2 with optimisation level 1 and a transpiler seed of 42; no provider or QPU was contacted. These are local resource estimates, not a scalability theorem, a hardware execution result, or a deployment-readiness guarantee.

The frozen unitary arm is not the shallowest arm in this local comparison; the no-pooling and measurement-style arms have lower logical depth, while the coherent extension and SU(4) arm add different forms of expressivity. The engineering distinction is instead that the frozen arm avoids the measurement and classical branch in the pooling stage. Its transpiled depth is higher than the measurement arm under this particular target and layout, which is a useful warning against treating removal of dynamic control as an automatic reduction in compiled depth. A different compiler, coupling map, or native instruction set could change that ordering; no such hardware-dependent conclusion is made here.

The finite trainability audit samples 200 random parameter initialisations and four random inputs per size on “default.qubit” with backpropagation. For a designated live slot, the observed gradient variances at  $n = 4, 6, 8, 10, 12, 14$  are respectively 0.0134, 0.0093, 0.0044, 0.0019, 0.00073, 0.00104; an exponential fit has  $R^2 = 0.923$  for that designated slot, whereas the median-live-slot fit prefers a power law with  $R^2 = 0.663$  and is non-monotone at  $n = 14$ . This finite diagnostic motivates caution about trainability; it is not evidence of a universal barren-plateau law or its absence.

### J. Finite trainability, expressibility, and Lie-algebra diagnostics

We report the model-analysis evidence in enough detail to show both what was measured and what the measurements cannot establish. These diagnostics concern the frozen ansatz and its finite-size instances, not the four-task test scores. They are included because parameter count, state-space coverage, and gradients answer different questions in a variational circuit.

TABLE XVI

LOCAL NOISE-LADDER VALIDATION ACCURACY AVERAGED OVER THREE CLEAN CHECKPOINTS AND 25 VALIDATION EXAMPLES PER CHECKPOINT. THE ZERO-NOISE COLUMN IS AN EXACT STATEVECTOR REFERENCE; NONZERO COLUMNS USE 128-SHOT LOCAL AER SIMULATION. NON-MONOTONE VALUES ARE RETAINED BECAUSE THE EXPERIMENT WAS NOT RETRAINED AND IS NOT A DEVICE-NOISE BENCHMARK.

| Task                 | Channel      | $p = 0$ | $p = 0.005$ | $p = 0.01$ | $p = 0.02$ |
|----------------------|--------------|---------|-------------|------------|------------|
| MNIST 3-vs-5         | Depolarising | 0.4400  | 0.4667      | 0.4933     | 0.5333     |
|                      | Dephasing    | 0.4400  | 0.5067      | 0.5067     | 0.4267     |
| Fashion-MNIST 0-vs-6 | Depolarising | 0.5867  | 0.4933      | 0.4667     | 0.4667     |
|                      | Dephasing    | 0.5867  | 0.5067      | 0.5333     | 0.4400     |

TABLE XVII

TOTAL CIRCUIT-DEPTH ESTIMATES FROM THE LOCAL FAKEPEEKSKILL TRANSPILATION AUDIT. COUNTS INCLUDE STATE PREPARATION, MODEL BODY, AND READOUT; NO HARDWARE EXECUTION IS IMPLIED.

| $n$ | Logical | Decomposed | Transpiled |
|-----|---------|------------|------------|
| 4   | 38      | 73         | 343        |
| 6   | 65      | 191        | 934        |
| 8   | 113     | 626        | 2,549      |
| 10  | 116     | 2,160      | 7,362      |

1) *Gradient-variance sweep*: For each  $n \in \{4, 6, 8, 10, 12, 14\}$ , we drew 200 parameter initialisations and four random encoded inputs, yielding 800 gradient draws per size. Angles were sampled uniformly from  $[0, 2\pi)$  and gradients were computed with backpropagation on `default.qubit`. A designated convolution-slot index was kept fixed across sizes, and a second summary took the median variance over slots that were live under the finite diagnostic. The designated slot is the more stable cross-size estimand; the live-slot population changes with the active-wire schedule and therefore makes its median a descriptive secondary reduction.

The controlled-slot series is compatible with either an exponential fit,  $0.0466 \exp(-0.3035n)$  with  $R^2 = 0.923$ , or a power-law fit,  $0.5162n^{-2.430}$  with  $R^2 = 0.915$ , on only six points. The difference in fit quality is too small to identify a decay law, and the designated variance increases at  $n = 14$  relative to  $n = 12$ . The median-live-slot series is even less suitable for extrapolation because the number and composition of live slots are non-monotone. The defensible statement is therefore finite: gradients remain measurable in the tested range and no barren plateau is observed through  $n = 14$  under this protocol. Neither the presence nor absence of an asymptotic barren plateau follows.

The  $n = 10$  row independently reproduces the 74-slot effective count obtained from the archived-weight, fixed-input audit, despite using random parameters, random inputs, and a variance criterion. This agreement strengthens the interpretation that the distinction between allocated and effective slots is a property of the executed topology. It does not mean that 74 is an effective dimension in the statistical-learning sense.

2) *Dynamical Lie algebra*: We computed commutator closure in the Pauli basis for three generator sets: “parameterized” contains the trainable gate generators as they appear in the

ansatz; “propagated” conjugates those generators through the fixed prefix so that it describes the ordered circuit factorisation; and “full” includes the fixed entangling gates as well. The latter two are the relevant sets when asking whether the complete circuit is contained in the exponential of a small algebra. The exact special-unitary ceiling is  $4^n - 1$ .

At the headline  $n = 10$ , the propagated and full closures are the complete  $\mathfrak{su}(2^{10})$  algebra of dimension 1,048,575. The literal parameterized set has dimension 65,550 because its connectivity decomposes into full blocks, but this smaller number is not the algebra relevant to the complete ordered circuit’s fixed entanglers. The exact full closure is a negative result for the planned polynomial-size DLA trainability certificate. It is also not a hardness theorem: the tape is shallow, the headline state has only 1,024 amplitudes, and other simulation methods may remain effective at larger sizes.

The closure routine was checked against controls with independently known dimensions: local rotations give  $3n$ , and the open-chain transverse-field Ising control gives  $2n^2 - n$ . Both controls agree with their expected values over the tested sizes. This makes the FQCNN result interpretable as a property of the ansatz rather than an artefact of a closure routine that always returns the full algebra. The input-dependent amplitude-embedding operation is not included in the generator set; it defines the state on which the variational ansatz acts.

3) *Expressibility and entangling capability*: For the headline ten-qubit ansatz, one fixed amplitude-encoded input was held constant while 3,000 parameter pairs were sampled. The fidelity distribution between each pair of output states was compared with the finite-sample Haar fidelity reference using 75 bins. This is an ansatz-ensemble diagnostic, not a measure of variation across the image dataset. The resulting KL divergence is  $1.087 \times 10^{-6}$ , while the Haar reference scored with the same sample size and bins has the same value to the reported precision. The mean pairwise fidelity is  $9.742 \times 10^{-4}$  versus the Haar expectation  $9.766 \times 10^{-4}$ . The Meyer–Wallach entangling capability is  $0.9745 \pm 0.0135$  over 3,000 states, with observed range 0.895–0.995.

The ansatz is therefore Haar-indistinguishable at this resolution, rather than “maximally expressible” in an unrestricted sense. The controls show that the metric can detect gross deviations, but the Haar law places 0.9999989 of its mass in the first of these 75 bins at ten qubits, which limits the resolution. The result should be read alongside the gradient sweep: high expressibility under one finite fidelity statistic

TABLE XVIII

POOLING-ARM PRACTICALITY COMPARISON AT  $n = 4$  ON THE SAME LOCAL FAKEPEEKSKILL BACKENDV2 TARGET. ENTRIES ARE DEPTH/OPERATION COUNT AT LOGICAL, DECOMPOSED, AND TRANSPILED LEVELS; THEY ARE LOCAL RESOURCE ESTIMATES, NOT HARDWARE EXECUTIONS.

| Arm         | Mode        | Logical | Decomposed | Transpiled | Control characteristic                  |
|-------------|-------------|---------|------------|------------|-----------------------------------------|
| No pooling  | none        | 30/78   | 30/78      | 225/381    | Wires retired without transfer          |
| Measurement | measurement | 36/87   | 36/87      | 240/401    | 3 measurements and 3 classical branches |
| Unitary     | unitary     | 36/90   | 48/108     | 280/457    | CRY/CRZ; no dynamic branch              |
| Coherent    | coherent    | 38/90   | 54/117     | 309/493    | Additional keep-to-discard CRY          |
| SU(4)       | su4         | 32/81   | 44/111     | 278/475    | General two-qubit unitary arm           |

TABLE XIX

FINITE GRADIENT-VARIANCE DIAGNOSTICS FOR THE FROZEN ANSATZ FAMILY. “LIVE” IS THE NUMBER OF SLOTS RETAINED BY THE  $10^{-12}$  FINITE CRITERION. THE DESIGNATED SLOT HAS THE SAME INDEX AND CONVOLUTIONAL ROLE AT EVERY SIZE; THE MEDIAN IS NOT USED TO CLAIM AN ASYMPTOTIC LAW.

| $n$ | Allocated slots | Live slots | Median live variance   | Designated variance    | Mean $ \partial f / \partial \theta $ |
|-----|-----------------|------------|------------------------|------------------------|---------------------------------------|
| 4   | 188             | 62         | $2.816 \times 10^{-2}$ | $1.335 \times 10^{-2}$ | $1.287 \times 10^{-1}$                |
| 6   | 194             | 62         | $1.413 \times 10^{-2}$ | $9.270 \times 10^{-3}$ | $8.788 \times 10^{-2}$                |
| 8   | 260             | 122        | $2.910 \times 10^{-3}$ | $4.416 \times 10^{-3}$ | $4.653 \times 10^{-2}$                |
| 10  | 269             | 74         | $3.768 \times 10^{-3}$ | $1.889 \times 10^{-3}$ | $4.092 \times 10^{-2}$                |
| 12  | 278             | 128        | $3.334 \times 10^{-4}$ | $7.304 \times 10^{-4}$ | $1.619 \times 10^{-2}$                |
| 14  | 287             | 74         | $2.951 \times 10^{-3}$ | $1.037 \times 10^{-3}$ | $3.352 \times 10^{-2}$                |

TABLE XX

DYNAMICAL LIE-ALGEBRA CLOSURE OF THE FROZEN ANSATZ. EXACT ROWS ENUMERATE THE CLOSURE;  $\geq 120,000$  MARKS THE EXPLICIT ENUMERATION CAP FOR THE LARGER INSTANCES AND IS A LOWER BOUND, NOT AN EQUALITY.

| $n$ | $4^n - 1$   | Parameterized | Propagated     | Full           | Status             |
|-----|-------------|---------------|----------------|----------------|--------------------|
| 4   | 255         | 255           | 255            | 255            | exact closure      |
| 6   | 4,095       | 270           | 4,095          | 4,095          | exact closure      |
| 8   | 65,535      | 65,535        | 65,535         | 65,535         | exact closure      |
| 10  | 1,048,575   | 65,550        | 1,048,575      | 1,048,575      | exact closure      |
| 12  | 16,777,215  | 65,790        | $\geq 120,000$ | $\geq 120,000$ | capped lower bound |
| 14  | 268,435,455 | 65,805        | $\geq 120,000$ | $\geq 120,000$ | capped lower bound |

TABLE XXI

CONTROLS FOR INTERPRETING THE FINITE EXPRESSIBILITY SCORE AT  $n = 10$ . A SMALL KL VALUE IS MEANINGFUL ONLY RELATIVE TO THE IDENTICALLY SCORED HAAR REFERENCE AND THE DEVIATION CONTROLS.

| Ensemble or control                | KL divergence          |
|------------------------------------|------------------------|
| Frozen ansatz                      | $1.087 \times 10^{-6}$ |
| Haar reference                     | $1.087 \times 10^{-6}$ |
| Ten-times inflated Haar fidelities | 3.823                  |
| Degenerate fidelity ensemble       | 25.809                 |

TABLE XXII

CAPACITY-SCALING READINGS AT THE HISTORICAL HEADLINE TRAINING SIZE  $N = 7,599$ . DIFFERENT CHOICES OF  $T$  ANSWER DIFFERENT ACCOUNTING QUESTIONS.

| Reading of $T$                  | $T$ | $g(T, N)$ |
|---------------------------------|-----|-----------|
| Trainable gates on tape         | 222 | 0.3973    |
| Gates driven by effective slots | 218 | 0.3930    |
| Effective parameter slots       | 74  | 0.2047    |
| Allocated parameter slots       | 269 | 0.4450    |

does not by itself determine the gradients of a shallow, structured parameterisation.

4) *Capacity-scaling diagnostic*: Finally, we evaluated the finite gate-count expression  $g(T, N) = \sqrt{T \log(T)/N}$  on the historical headline training partition, where  $N = 7,599$ . The result is shown in Table XXII. Here  $T$  must count trainable gate occurrences rather than parameter slots. The quantity is reported as a scaling diagnostic with an omitted theorem-dependent constant; it is not presented as a numerical generalisation guarantee.

All four values are below one for this finite reading, but the effective-slot value must not be called the bound: 74 slots drive 218 trainable gate occurrences, while 191 allocated slots never reach the tape. The gate-based readings 0.3973 and 0.3930 differ by only about one percent. The parameter audit can still support a statement about allocation efficiency, but it cannot be substituted into a gate-count theorem to manufacture a stronger generalisation conclusion.

5) *Joint interpretation*: The three diagnostics expose a useful tension. The state ensemble is close to the finite Haar

reference, the complete ordered ansatz reaches the full special-unitary algebra at ten qubits, and yet the designated gradient remains measurable through the tested range. The correct resolution is not to select the most favourable interpretation. The expressibility statistic has finite resolution, the DLA result removes rather than supplies a polynomial-size certificate, and the gradient sweep has too few sizes to identify an asymptotic law. Together they support an empirical statement about the tested architecture and sizes, with no universal trainability claim in either direction.

## VII. DISCUSSION

This paper separates architectural identity, statistical behavior, information dynamics, and implementation cost. This separation is important because those questions can point in different directions. An exact channel equivalence does not imply an accuracy advantage, and a finite trainability diagnostic does not imply scalable optimization.

### A. Architectural contribution versus predictive performance

The central contribution is a change in the control boundary of pooling. The controlled unitary followed by a partial trace implements the same measure-and-condition channel as the compared mid-circuit construction. Thus, the coherent block removes intermediate measurement, reset, and classical feed-forward from the frozen path while preserving the corresponding retained register statistics. It does not recover the discarded subsystem, and the partial trace remains part of the reduced-state semantics. The E1, E2, and E3 results jointly make this distinction testable at fixed parameters and after training.

Predictive performance tells a different story. In the primary four-task matrix, the FQCNN is below both classical baselines and above TTN in mean accuracy on every task, but none of the twelve paired effects survives Holm correction. The secondary pooling campaign shows that removing pooling can hurt, while a larger SU(4) pooling family has finite headroom over the frozen three-angle block. These results do not weaken the architectural theorem; they prevent it from being used as a proxy for model superiority. The appropriate claim is therefore that the paper supplies a coherent implementation with measured, task-dependent predictive behavior and an explicit family-level ablation boundary.

### B. What the diagnostics reveal about the frozen circuit

The information audit explains why a globally coherent computation should not be described as locally information-preserving at every stage. The full statevector remains pure under the unitary path, whereas the active register becomes mixed through entanglement with wires that are no longer active. The reported keep–discard mutual information decreases sharply across the cascade on the archived inputs. This is compatible with dimensional reduction, but it does not establish that each pooling stage concentrates the task-relevant feature or that the final stage is useful on every task.

The finite trainability results impose a second boundary. Measurable gradients through  $n = 14$  do not identify an

asymptotic scaling law; the full special-unitary Lie algebra at  $n = 10$  rules out a polynomial-DLA certificate for the complete frozen tape; and the expressibility statistic is a finite-resolution comparison rather than a proof of maximal expressivity. The generalization expression is reported as a capacity diagnostic and is not a generalization guarantee. Taken together, these checks favour a mechanism-oriented interpretation over a single favorable slogan.

### C. Implementation implications and scaling boundary

The resource audit makes the cost of the coherent path concrete. On the local FakePeekskill transpilation target, total transpiled depth and operation count increase from 343 and 567 at four qubits to 7,362 and 15,239 at ten qubits. The pooling-arm audit also shows that a unitary replacement is not automatically the shallowest implementation: the measurement, unitary, coherent, and SU(4) variants have different logical and transpiled costs. These are compiler- and backend-specific observations, not device benchmarks.

The ten-qubit experiment is consequently a controlled validation scale, not a claim of practical deployment at arbitrary register size. Global amplitude encoding requires a state-preparation interface that is itself outside the learned variational tape, and all reported numerical results were produced by classical simulation or a local fake backend. A hardware study would need to declare the provider, calibration date, shot budget, compilation settings, state-preparation method, and error model. Those external actions are not required to establish the present architectural result and remain outside this release.

### D. Reproducibility as part of the contribution

The primary comparison binds every arm to the same ordered sample identifiers, split manifests, five seeds, validation protocol, and final test partition. The evidence files preserve per-seed outcomes rather than only aggregate means, and the source revision, environment locks, figures, and validation reports are packaged together. This makes the negative comparisons and the positive pooling checks auditable. It also makes the scope of replication clear: a new implementation should first reproduce the frozen simulator artifacts, then extend the task or hardware domain with a separately registered protocol.

### E. Limitations and Future Work

The current evaluation is limited to four binary tasks drawn from three  $28 \times 28$  grayscale dataset families, small training subsets, five-seed comparisons, finite-size diagnostics, and classical simulation/local noise models. The globally amplitude-encoded register is not an image-space convolutional representation, and the experiments do not establish quantum advantage, asymptotic trainability, noise robustness on a device, or real-QPU performance. Scaling to larger datasets, multi-class tasks, alternative encodings, and optional hardware validation remains outside this release.

## VIII. NOVELTY AND TECHNICAL DIFFERENTIATION

The proposed FQCNN departs from previous quantum convolutional architectures on four precise technical fronts that together make up the main contributions of this paper.

### A. Fully unitary pipeline

The canonical QCNN construction relies on mid-circuit measurement and classical post-processing at pooling [10], which requires dynamic-circuit capability at each pooling layer. Theorem 1 shows that the coherent block reproduces that construction exactly on the retained register, so the contribution here is the removal of that requirement rather than the recovery of lost information. Here, encoding, convolution, pooling, and classification are merged into a single unitary transform with no intermediate measurements. This is an architectural compatibility distinction; whether it improves an implementation depends on the target device, compiler, and control stack, and is not established by the simulator-only study here.

### B. Coherent unitary pooling

Instead of a mid-circuit projection, which incurs latency and reset overheads on physical devices and requires dynamic-circuit support, the architecture uses the measurement-free pooling block of (14): features are routed from the compressed sub-register into the retained register by controlled rotations, with the compressed qubit as control. After pooling, the compressed sub-register is excluded from the remainder of the circuit, which is equivalent to a partial trace over those subsystems.

The contribution is precisely delimited by Section IV-F. Theorem 1 establishes that this block reproduces the corresponding measure-and-condition channel exactly, so it is not more expressive than measurement-based pooling; Proposition 1 establishes that the wider unitary family strictly contains that class, so the frozen block is one explicitly simulated member of that family. What the block delivers is the same channel without mid-circuit measurement, reset, or classical feed-forward, with the global register pure up to the single terminal measurement and the whole circuit differentiable by the adjoint method. The discard step is accounted for as a partial trace rather than as a unitary on the full register.

### C. Register-index all-to-all convolutional kernel

The kernel applies a fixed six-CNOT pattern in each  $2 \times 2$  register-index window and couples every pair within that window. The gate support is bounded and the per-window depth is fixed; because state preparation is global amplitude encoding, this should not be read as image-space locality or translational equivariance. Parameter sharing reduces allocation growth, while trainability is assessed only through the finite diagnostics reported in the evidence package, not inferred from locality alone.

### D. Global amplitude encoding and register-index structure

Inputs are min-max normalised, flattened, zero-padded to 1024 entries, and prepared by the single global amplitude-encoding operation described above. No local rotation map or data-dependent cyclic-ring correlation layer is part of the frozen main path. The subsequent variational kernel and pooling blocks use shared register-index supports; this distinction keeps the implementation and the paper’s locality language aligned.

## DATA AND CODE AVAILABILITY

The manuscript source, implementation, tests, environment locks, machine-readable evidence, figures, and reproduction instructions are maintained in the version-controlled repository at the release commit identified by the submission package manifest. That manifest records the source revision and SHA-256 hashes for the files used to produce the reported results. MNIST, Fashion-MNIST, and KMNIST are obtained from their upstream sources and are not redistributed here; the dataset-provenance artifact records the source URLs, labels, local snapshot identities, and checksums needed to reconstruct the registered tasks. The validated aggregates are simulator evidence, and ignored run weights and prediction archives are regenerable rather than a substitute for a public data deposit. A persistent repository DOI or immutable URL will be added to the final submission record after the venue is selected.

## IX. CONCLUSION

We proposed a fully quantum-native convolutional neural network (FQCNN) that performs state preparation, convolution, pooling, and classification through a single coherent unitary path before the terminal observable. The exact pooling result identifies removal of mid-circuit control as the architectural contribution; the Q1 aggregate and transfer experiments quantify empirical behavior against same-split baselines without presupposing an advantage. Resource, noise, and trainability statements are restricted to their finite local evidence and simulation scope. Future work may extend the evaluation to larger and multi-class tasks or optional real quantum hardware; neither extension is required for this release.

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